

ITC C106 — Machine Problems

Java Source Codes

Problem 1 — Lotto 6/42 (Arrays)

Allows the user to enter 6 numbers, generate 6 random winning numbers, and displays the prize based on the number of matches.

```
import java.util.Random;
import java.util.Scanner;

public class Lotto {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] bet = new int[6];
        int[] winning = new int[6];
        Random rand = new Random();

        System.out.println("=== LOTTO 6/42 ===");
        System.out.println("Enter your 6 numbers (1-42):");
        for (int i = 0; i < 6; i++) {
            System.out.print("Number " + (i + 1) + ": ");
            bet[i] = sc.nextInt();
        }

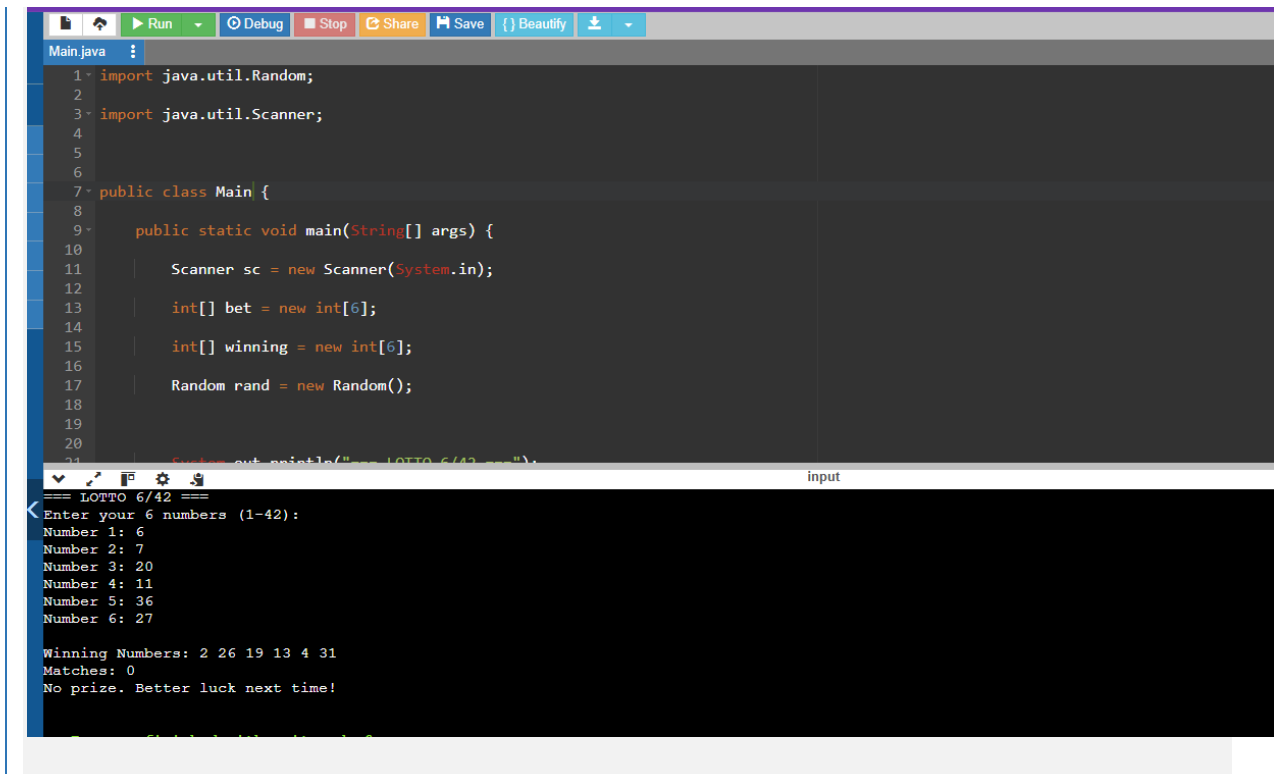
        for (int i = 0; i < 6; i++) {
            winning[i] = rand.nextInt(42) + 1;
        }

        System.out.print("\nWinning Numbers: ");
        for (int n : winning) System.out.print(n + " ");

        int matches = 0;
        for (int b : bet)
            for (int w : winning)
                if (b == w) { matches++; break; }

        System.out.println("\nMatches: " + matches);

        switch (matches) {
            case 6: System.out.println("Prize: P 31,621,223.97"); break;
            case 5: System.out.println("Prize: P 25,000.00"); break;
            case 4: System.out.println("Prize: P 1,000.00"); break;
            case 3: System.out.println("Prize: P 20.00"); break;
            default: System.out.println("No prize. Better luck next time!");
        }
        sc.close();
    }
}
```



The screenshot shows an IDE with a Java file named 'Main.java'. The code imports `java.util.Random` and `java.util.Scanner`. It defines a `Main` class with a `main` method. Inside `main`, it creates a `Scanner` for input, initializes arrays for bets and winning numbers, and creates a `Random` object. The output window shows the program's execution: it prints '=== LOTTO 6/42 ===', prompts for 6 numbers, reads the input (6, 7, 20, 11, 36, 27), displays the winning numbers (2, 26, 19, 13, 4, 31), shows 0 matches, and prints 'No prize. Better luck next time!'.

```
1 import java.util.Random;
2
3 import java.util.Scanner;
4
5
6
7 public class Main {
8
9     public static void main(String[] args) {
10
11         Scanner sc = new Scanner(System.in);
12
13         int[] bet = new int[6];
14
15         int[] winning = new int[6];
16
17         Random rand = new Random();
18
19
20
21         System.out.println("=== LOTTO 6/42 ===");
22
23         System.out.println("Enter your 6 numbers (1-42):");
24         for (int i = 0; i < bet.length; i++) {
25             bet[i] = sc.nextInt();
26         }
27
28         System.out.println("Winning Numbers: 2 26 19 13 4 31");
29         int matches = 0;
30         for (int i = 0; i < bet.length; i++) {
31             for (int j = 0; j < winning.length; j++) {
32                 if (bet[i] == winning[j]) {
33                     matches++;
34                 }
35             }
36         }
37
38         System.out.println("Matches: " + matches);
39         if (matches == 0) {
40             System.out.println("No prize. Better luck next time!");
41         }
42     }
43 }
```

input

```
=== LOTTO 6/42 ===
Enter your 6 numbers (1-42):
Number 1: 6
Number 2: 7
Number 3: 20
Number 4: 11
Number 5: 36
Number 6: 27

Winning Numbers: 2 26 19 13 4 31
Matches: 0
No prize. Better luck next time!
```

Problem 2 — BMI Calculator (Methods)

A void method that accepts height (cm) and weight (kg), computes BMI using the formula $(\text{weight} / \text{height}^2) * 10000$, and displays the category.

```
import java.util.Scanner;

public class Main {

    static void calculateBMI(double height, double weight) {

        double bmi = (weight / (height * height)) * 10000;

        System.out.printf("BMI: %.2f\n", bmi);

        if (bmi < 18.5)

            System.out.println("Category: Underweight");

        else if (bmi <= 24.9)

            System.out.println("Category: Normal weight");

        else if (bmi <= 29.9)

            System.out.println("Category: Overweight");

    }

}
```

```
        else

            System.out.println("Category: Obese");

    }

    public static void main(String[] args) {

        System.out.println("    BMI CALCULATOR    ");

        Scanner xx = new Scanner(System.in);

        System.out.print("Enter your height in centimeters: ");

        double height = xx.nextDouble();

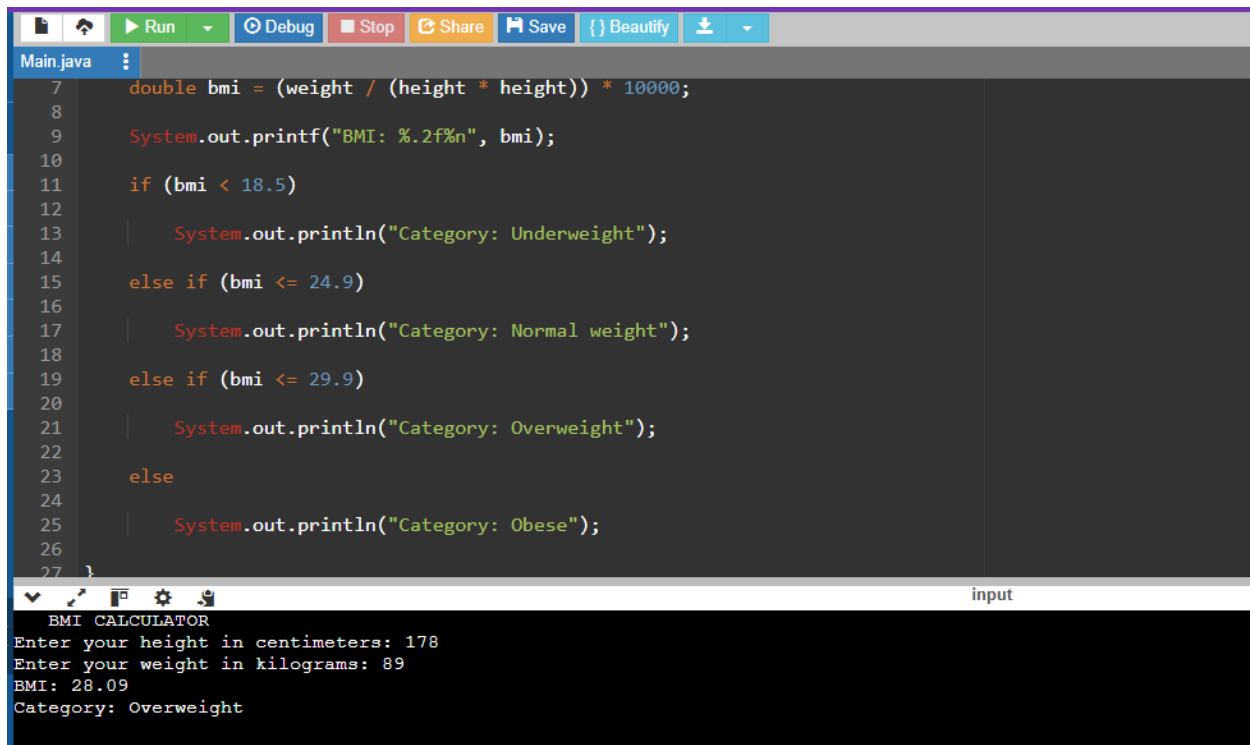
        System.out.print("Enter your weight in kilograms: ");

        double weight = xx.nextDouble();

        calculateBMI(height, weight);

    }

}
```



The screenshot shows an IDE window titled 'Main.java' with the following code:

```
7 double bmi = (weight / (height * height)) * 10000;  
8  
9 System.out.printf("BMI: %.2f%n", bmi);  
10  
11 if (bmi < 18.5)  
12 |     System.out.println("Category: Underweight");  
13  
14 else if (bmi <= 24.9)  
15 |     System.out.println("Category: Normal weight");  
16  
17 else if (bmi <= 29.9)  
18 |     System.out.println("Category: Overweight");  
19  
20 else  
21 |     System.out.println("Category: Obese");  
22  
23  
24  
25  
26  
27 }
```

Below the code editor, the output console shows the program's execution:

```
BMI CALCULATOR  
Enter your height in centimeters: 178  
Enter your weight in kilograms: 89  
BMI: 28.09  
Category: Overweight
```

Problem 3 — Arrays & Sorting

Stores user-inputted integers in an array and provides a menu to display, filter by index, or sort the values in ascending or descending order.

```
import java.util.Arrays;  
  
import java.util.Scanner;  
  
public class Main {  
  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
  
        System.out.println("=====");  
  
        System.out.println("      = ARRAY & SORTING =      ");  
  
        System.out.println("=====");  
  
  
        System.out.print("Enter number of integers: ");
```

```

int n = sc.nextInt();

int[] numbers = new int[n];

for (int i = 0; i < n; i++) {
    System.out.print("Enter number " + (i + 1) + ": ");
    numbers[i] = sc.nextInt();
}

String choice;

do {
    System.out.println();
    System.out.println("=====");
    System.out.println("= LIST OF CHOICES =");
    System.out.println("=====");
    System.out.println("1) Display the numbers");
    System.out.println("2) Display the values of even indexes");
    System.out.println("3) Display the values of odd indexes");
    System.out.println("4) Display the values in ascending order");
    System.out.println("5) Display the values in descending order");
    System.out.println("X) Exit program");
    System.out.println("-----");
    System.out.print("Enter your choice: ");

    choice = sc.next().toUpperCase();

    System.out.println("-----");

    switch (choice) {
        case "1":
            System.out.print("Numbers: ");

            for (int i = 0; i < numbers.length; i++) {
                System.out.print(numbers[i] + " ");
            }

```

```
}

System.out.println();

break;

case "2":

System.out.print("Even indexes: ");

for (int i = 0; i < numbers.length; i += 2) {

System.out.print(numbers[i] + " ");

}

System.out.println();

break;

case "3":

System.out.print("Odd indexes: ");

for (int i = 1; i < numbers.length; i += 2) {

System.out.print(numbers[i] + " ");

}

System.out.println();

break;

case "4":

int[] ascending = Arrays.copyOf(numbers, numbers.length);

Arrays.sort(ascending);

System.out.print("Ascending order: ");

for (int i = 0; i < ascending.length; i++) {

System.out.print(ascending[i] + " ");

}

System.out.println();

break;

case "5":

int[] descending = Arrays.copyOf(numbers, numbers.length);

Arrays.sort(descending);
```

```
System.out.print("Descending order: ");

for (int i = descending.length - 1; i >= 0; i--) {

System.out.print(descending[i] + " ");

}

System.out.println();

break;

case "X":

System.out.println("=====");

System.out.println(" Exiting program. ");

System.out.println(" Thank you! Bye! ");

System.out.println("=====");

break;


default:

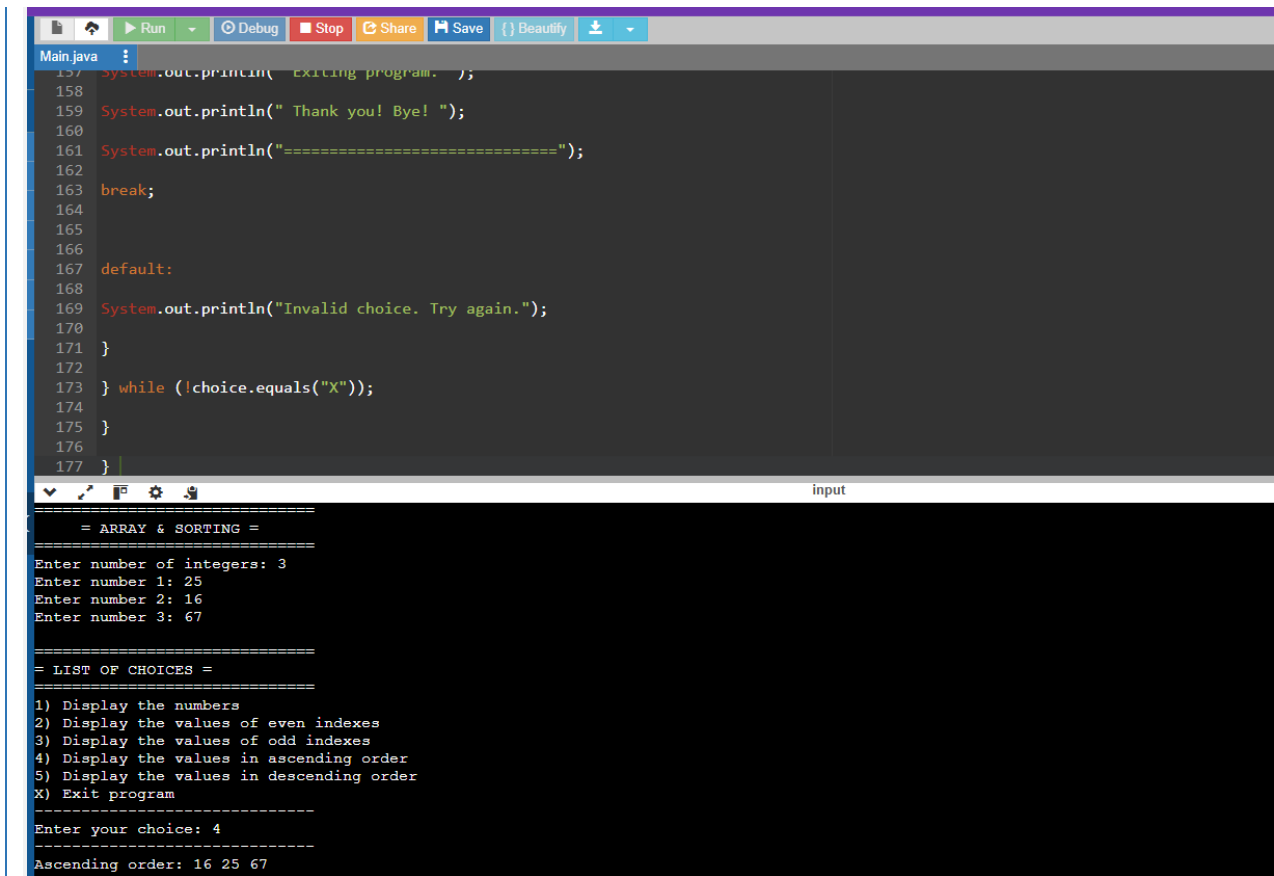
System.out.println("Invalid choice. Try again.");

}

} while (!choice.equals("X"));

}

}
```



The screenshot shows an IDE with a Java file named 'Main.java'. The code is a menu-driven program for array operations. It prompts the user to enter the number of integers and the integers themselves. It then displays a menu of choices: 1) Display the numbers, 2) Display the values of even indexes, 3) Display the values of odd indexes, 4) Display the values in ascending order, 5) Display the values in descending order, and X) Exit program. The user has entered '4', and the output shows the numbers in ascending order: 16 25 67.

```
157 System.out.println("Exiting program. ");
158
159 System.out.println(" Thank you! Bye! ");
160
161 System.out.println("=====");
162
163 break;
164
165
166 default:
167
168 System.out.println("Invalid choice. Try again.");
169
170 }
171
172 } while (!choice.equals("X"));
173
174 }
175
176
177 }
```

Input

```
=====
= ARRAY & SORTING =
=====
Enter number of integers: 3
Enter number 1: 25
Enter number 2: 16
Enter number 3: 67
=====
= LIST OF CHOICES =
=====
1) Display the numbers
2) Display the values of even indexes
3) Display the values of odd indexes
4) Display the values in ascending order
5) Display the values in descending order
X) Exit program
=====
Enter your choice: 4
=====
Ascending order: 16 25 67
```

Problem 4 — Stacks & Queues

Converts a string to a character ArrayList and supports both stack (LIFO) and queue (FIFO) operations through a menu-driven interface.

```
import java.util.ArrayList;
import java.util.Scanner;

public class StackQueue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter any text: ");
        String text = sc.nextLine();
        ArrayList<Character> arr = new ArrayList<>();
        for (char c : text.toCharArray()) arr.add(c);

        String choice;
        do {
            System.out.println("\nA. Display the string");
            System.out.println("B. Pop (remove last)");
            System.out.println("C. Top (peek last)");
            System.out.println("D. Push/Enqueue (add character)");
            System.out.println("E. Dequeue (remove first)");
```



```

        System.out.println("F. Front (peek first)");
        System.out.print("Enter a choice [X to exit]: ");
        choice = sc.nextLine().trim();

        switch (choice.toUpperCase()) {
            case "A":
                if (arr.isEmpty()) System.out.println("empty");
                else { for (char c : arr) System.out.print(c);
System.out.println(); }
                break;
            case "B":
                if (arr.isEmpty()) System.out.println("empty");
                else { arr.remove(arr.size() - 1);
System.out.println("Character Popped."); }
                break;
            case "C":
                if (arr.isEmpty()) System.out.println("empty");
                else System.out.println("Top character is " +
arr.get(arr.size() - 1));
                break;
            case "D":
                System.out.print("Enter a character: ");
                char ch = sc.nextLine().charAt(0);
                arr.add(ch);
                System.out.println("Character Pushed/Enqueued.");
                break;
            case "E":
                if (arr.isEmpty()) System.out.println("empty");
                else { arr.remove(0); System.out.println("Character
Dequeued."); }
                break;
            case "F":
                if (arr.isEmpty()) System.out.println("empty");
                else System.out.println("Front Character is " + arr.get(0));
                break;
            case "X":
                System.out.println("End"); break;
            default:
                System.out.println("Invalid choice.");
        }
    } while (!choice.equalsIgnoreCase("X"));
}
}

```